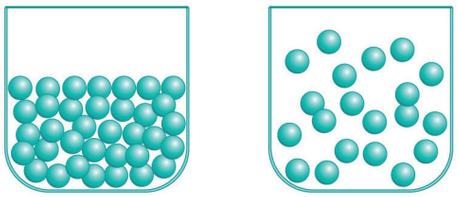
14.1 Means of Transport in Plants

The transport of materials in and out of the cells is carried out by diffusion and active transport in plants.

14.1.1 Diffusion

The movement of solid liquid and gaseous molecules from a region of higher concentration to a region of their lower concentration without the utilization of energy is called **diffusion**. This is a passive process.



Before diffusion After diffusion

Figure 14.1 Diffusion

14.1.2 Active Transport

Active transport utilizes energy to pump molecules against a concentration gradient. Active transport is carried out by membrane bound proteins. These proteins use energy to carry substances across the cell membrane hence they are often referred to as **pumps**. These pumps can transport substances from a low concentration to a high concentration (**'uphill' transport**).

14.1.3 Osmosis

Osmosis is the **movement of solvent** or water molecules from the **region of higher concentration** to the region of lower concentration through a semi-permeable membrane. This process is carried out till an equilibrium is reached. Osmosis is the passive movement of water or any other solvent molecules.

Plasmolysis

It occurs when a living plant cell is placed in a hypertonic solution water molecule moves out of the cell and resulting in the shrinkage of protoplasm away from the cell wall.

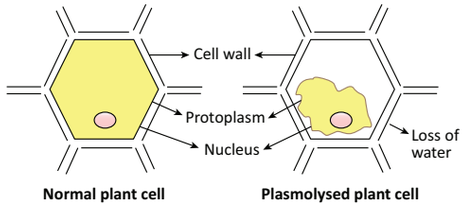
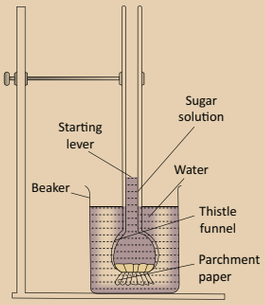


Figure 14.2 Plasmolysis

Activity 1

Demonstration of Osmosis

A thistle funnel whose mouth is covered with a semipermeable membrane, is filled with sucrose solution. It is kept inverted in a beaker containing water. The water will diffuse across the membrane due to osmosis and raise the level of the solution in the funnel.



Imbibition

Imbibition is a type of diffusion in which a solid plant material absorbs water and gets swelled up. eg. absorption of water by dry seeds and grapes. If it were not for imbibition, seedlings would not have been able to emerge out of the soil.

14.2 Root Hair-Water Absorbing Unit

There are millions of root hairs on the tip of the root which absorb water and minerals by diffusion. Root hairs are thin walled, slender extension of epidermal cell that increase the surface area of absorption.

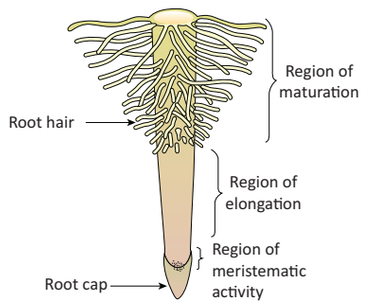


Figure 14.3 Root Tip with Root Hairs

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